Bayesian Hierarchical Model for $PM_{2.5}$ levels in Italy

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29-04-2026

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1 Introduction

Air pollution represents a major environmental and public health concern. Specifically, fine particulate matter $PM_{2.5}$ is among the most harmful pollutants due to its ability to penetrate deep into the respiratory system and its strong association with cardiovascular and pulmonary diseases.

A key challenge in modeling $PM_{2.5}$ concentrations lies in the presence of substantial heterogeneity across monitoring stations and irregular temporal coverage. Stations differ significantly in terms of geographical characteristics, local emission sources, and environmental conditions, while observational data often exhibit an unbalanced panel structure, with varying numbers of observations across stations and missing time periods.

In this context, Bayesian hierarchical models provide a natural and flexible framework to account for these complexities. By introducing group-level random effects, these models allow for partial pooling, enabling information sharing across stations while preserving local heterogeneity. Moreover, the Bayesian approach provides a coherent framework for uncertainty quantification and prior specification.

This paper develops a Bayesian hierarchical model for daily $PM_{2.5}$ concentrations observed across monitoring stations in Italy. Posterior inference is carried out using a Metropolis-within-Gibbs sampler. The analysis focuses on model specification, MCMC implementation, convergence diagnostics, sensitivity analysis, and posterior predictive checks.

2 Data

The dataset comprises daily $PM_{2.5}$ concentration measurements collected across monitoring stations in Italy during 2025. The raw data were sourced from the European Environment Agency (EEA) Air Quality portal, specifically via the EEA Air Quality e-Reporting, with metadata and station specifications validated through the EEA Air Quality Viewer.[1] After preprocessing and quality control, the final dataset contains 31,349 observations distributed across 116 stations.

The empirical distribution of the raw outcome variable is highly right-skewed, with a skewness of approximately 17.1. This strong asymmetry is primarily driven by extreme pollution events in the upper tail. To stabilize the variance and satisfy the distributional assumptions of the proposed model, a logarithmic transformation was applied. This transformation substantially reduces skewness (to approximately -0.29), yielding a more symmetric distribution (visible in Figure 1) and making the Gaussian likelihood a reasonable modeling choice.

$$y_{it} = \log(\mathrm{PM}_{2.5,it})$$

where y_{it} represents the transformed concentration for station i at time t. In what follows, each observation will be indexed by a single index i, with the mapping between observations and station-time pairs left implicit.

[1] EEA Air Quality e-Reporting: https://eeadmz1-downloads-webapp.azurewebsites.net/. EEA Air Quality Viewer: https://discomap.eea.europa.eu/App/AQViewer/index.html?fqn=Airquality_Dissem.b2g.measurements.

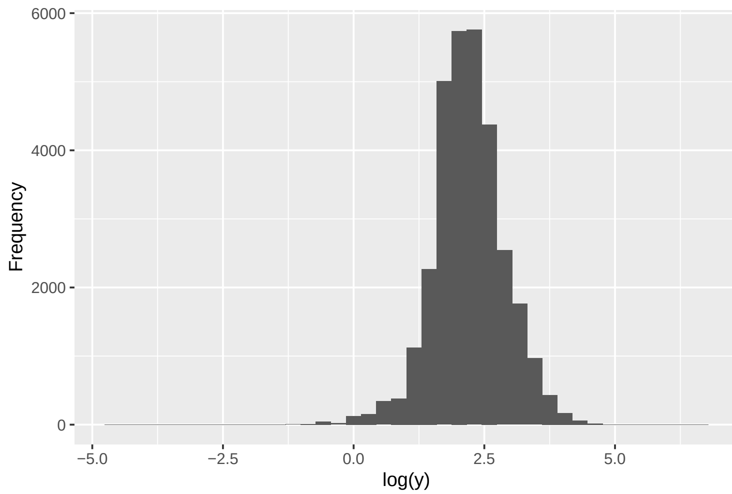

Figure 1: Distribution of the log-transformed PM2.5 concentrations.

The panel structure is markedly unbalanced, a common feature in environmental monitoring networks. The number of observations per station ranges from 69 to 332, with a median of approximately 299. Temporal coverage is also uneven across stations. While most stations provide observations for most of the year, 32 stations lack at least one full month of data, and for a few stations coverage drops to only 4 months. This heterogeneity is summarized in Figure 2, which reports both the distribution of the number of observations per station and the distribution of the number of months observed per station. Such an irregular structure poses a challenge for standard time-series methods.

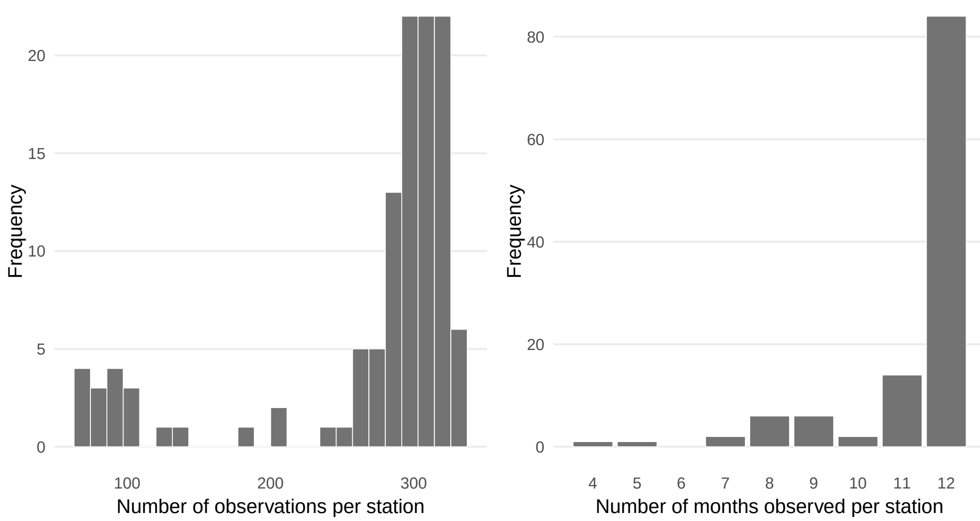

Figure 2: Data availability by station.

For the estimation of the Bayesian hierarchical model, only a restricted set of variables was retained from the cleaned dataset. The modelling dataset includes the log-transformed outcome variable, the station identifier, the month, and the day of the week. Observations with missing values in any of these variables were excluded from the modelling dataset, resulting in a complete-case sample. The variables `station_id`, `month`, and `day_of_week` were then treated as categorical factors.

Formally, the model is estimated on a dataset consisting of the variables $\{y_{it}, \text{station}_i, \text{month}_t, \text{day\_of\_week}_t\}$.

The corresponding design matrix includes an intercept, monthly indicators, and day-of-week indicators, while the station identifier is used to define the hierarchical grouping structure.

3 Model

The hierarchical nature of the resulting complete-case dataset, together with the strong imbalance in the panel structure, naturally motivates a multilevel modeling approach. In particular, the uneven number of observations across stations and the presence of incomplete temporal coverage make standard time-series or pooled regression models potentially inefficient and biased, as they are unable to properly account for unobserved heterogeneity across monitoring sites.

To address these issues, a Bayesian hierarchical specification is adopted, which allows for partial pooling across stations. This framework enables the model to borrow strength from the entire dataset while retaining station-specific components, thus providing robust inference even in the presence of sparse observations for some units.

A hierarchical model is considered for the log-transformed $\text{PM}_{2.5}$ concentrations. Let y_i denote the transformed outcome for observation i.

The covariates included in the model correspond to calendar effects. In particular, the fixed-effects design matrix is constructed as

$$X = \texttt{model.matrix}(\sim \text{month} + \text{day\_of\_week}),$$

so that the intercept is included by default together with monthly and day-of-week indicators. The resulting parametrization corresponds to a treatment coding scheme, in which the intercept represents the expected log $\text{PM}_{2.5}$ level for the reference categories (January and Monday), and the remaining coefficients measure deviations from these baselines. Station-level heterogeneity is not included among the fixed effects, but is instead modeled through station-specific random intercepts.

The likelihood is specified as:

$$y_i \mid \beta, \alpha, \sigma^2 \sim \mathcal{N}(x_i^\top \beta + \alpha_{j[i]}, \sigma^2)$$

where:

- x_i is the i-th row of the design matrix X,
- β is a vector of fixed-effect coefficients associated with the intercept, month indicators, and day-of-week indicators,
- $\alpha_{j[i]}$ is a station-specific random intercept accounting for time-invariant local characteristics.

3.1 Hierarchical structure

Station-specific effects are modeled as draws from a common Gaussian population distribution:

$$\alpha_j \mid \tau^2 \sim \mathcal{N}(0, \tau^2), \quad j = 1, \dots, J,$$

where τ^2 captures the cross-station variability in baseline pollution levels.

The assumption of a common Gaussian distribution reflects the idea that monitoring stations are exchangeable draws from a broader population of locations, once observed covariates are accounted for. This formulation allows the model to decompose variation in pollution levels into a systematic component (captured by $x_i^\top \beta$) and an unobserved station-specific component.

The hierarchical structure induces partial pooling: stations with limited observations are shrunk toward the overall mean, while stations with richer data retain more idiosyncratic estimates. This feature is particularly important in the present setting, where the panel is highly unbalanced and some stations provide only sparse temporal coverage. Shrinkage improves estimation stability and prevents overfitting to noisy local patterns.

3.2 Prior specification

Weakly informative priors are assigned to all parameters to regularize estimation while allowing the data to drive inference.

For the regression coefficients:

$$\beta \sim \mathcal{N}(b_0, B_0),$$

where $B_0 = cI$ is a diagonal covariance matrix. This specification reflects prior independence across coefficients and imposes mild shrinkage toward zero, helping to stabilize estimation in the presence of potentially correlated temporal indicators.

The observation variance is assigned the prior

$$\sigma^2 \sim \text{Inverse-Gamma}(a_\sigma, b_\sigma),$$

which provides a conjugate prior facilitating posterior computation while remaining weakly informative for suitably chosen hyperparameters.

For the group-level standard deviation:

$$\tau \sim \text{Half-Cauchy}(0, A).$$

The use of a half-Cauchy prior for τ avoids the well-known sensitivity of inverse-gamma priors for variance components, while providing weakly informative regularization.

3.3 Posterior inference

Posterior inference is performed via a Metropolis-within-Gibbs sampler. Conditional on the variance components, the regression coefficients β and the station-specific effects α admit Gaussian full conditional distributions and are therefore updated through Gibbs steps. The observation variance σ^2 is sampled from its inverse-gamma full conditional distribution.

The group-level scale parameter τ, for which no closed-form full conditional is available under the half-Cauchy prior, is updated through a random-walk Metropolis step. This hybrid scheme combines the computational convenience of conjugate updates with the flexibility required for the non-conjugate variance component.

The algorithm therefore alternates between Gibbs updates for β, α, and σ^2, and a Metropolis-Hastings update for τ.

3.4 Implementation details

Hyperparameters are selected to induce weakly informative priors. For the regression coefficients, the prior covariance matrix is set to $B_0 = cI$, with baseline choice $c = 100$, implying a large prior variance and only mild shrinkage toward zero. For the observation variance, the inverse-gamma hyperparameters are fixed at $a_\sigma = 2$ and $b_\sigma = 0.5$, yielding a diffuse prior. The value $b_\sigma = 0.5$ is chosen to be consistent with the empirical scale of the log-transformed outcome (whose sample variance is approximately 0.47), while remaining sufficiently weakly informative. The scale parameter of the half-Cauchy prior on τ is set to $A = 5$, allowing for substantial between-station variability while still providing regularization. The robustness of posterior inference to alternative hyperparameter choices is examined through a dedicated sensitivity analysis reported below.

Posterior inference is performed using a Metropolis-within-Gibbs sampler run for 5,000 iterations. Burn-in and thinning are applied as post-processing steps: the first 200 iterations are discarded and every 5th remaining draw is retained.

The Metropolis step for τ is implemented as a Gaussian random walk on the log scale. A tuning exercise is conducted to calibrate the proposal variance. With a proposal standard deviation of 0.15, the acceptance rate is approximately 0.48, while increasing it to 0.20 reduces the rate to approximately 0.38. Since posterior summaries and diagnostic plots are essentially unaffected, the larger proposal scale is retained to improve exploration of the parameter space.

Finally, convergence is assessed using both graphical (trace plots and autocorrelation functions) and numerical diagnostics, including the Geweke statistic and the effective sample size.

3.4.1 Sensitivity to prior specification

To assess whether posterior inference is materially affected by the baseline prior calibration, a one-at-a-time sensitivity analysis is performed. Starting from the baseline configuration $c = 100$, $A_\tau = 5$, and $b_\sigma = 0.5$, each hyperparameter is varied separately while keeping the others fixed.

Three prior components are examined. First, the prior variance of the regression coefficients is changed by setting $c = 10$ and $c = 1000$. Second, the scale parameter of the half-Cauchy prior on the station-level standard deviation is varied from $A_\tau = 5$ to $A_\tau = 2.5$ and $A_\tau = 10$. Third, the inverse-gamma scale parameter for the observation variance is increased from $b_\sigma = 0.5$ to $b_\sigma = 1$, while keeping $a_\sigma = 2$ fixed.

As reported in Table 1, posterior summaries are stable across all specifications. The posterior means of the intercept, σ^2, and τ remain virtually unchanged, and the substantive interpretation of the temporal effects and station-level heterogeneity is unaffected. Acceptance rates for the Metropolis step on τ also remain in a satisfactory range across specifications.

Table 1: Sensitivity analysis under alternative prior specifications.

Scenario	c	A_τ	b_σ	β_0	σ^2	τ	Acceptance (τ)
Baseline	100	5.0	0.5	2.603	0.325	0.299	0.388
Smaller prior variance for beta	10	5.0	0.5	2.602	0.325	0.298	0.394
Larger prior variance for beta	1000	5.0	0.5	2.603	0.325	0.299	0.401
Smaller A tau	100	2.5	0.5	2.603	0.325	0.299	0.384
Larger A tau	100	10.0	0.5	2.603	0.325	0.299	0.388
Larger b sigma	100	5.0	1.0	2.603	0.325	0.298	0.393

Overall, the sensitivity analysis indicates that posterior inference is not materially driven by the specific prior calibration. Instead, the likelihood appears sufficiently informative for the parameters of interest.

4 Results

4.1 Convergence diagnostics

Trace plots of the post-burn-in chains for the variance components (Figure 3) show no evidence of non-stationarity or systematic trends. Both chains fluctuate around stable levels, indicating satisfactory convergence of the Metropolis-within-Gibbs sampler.

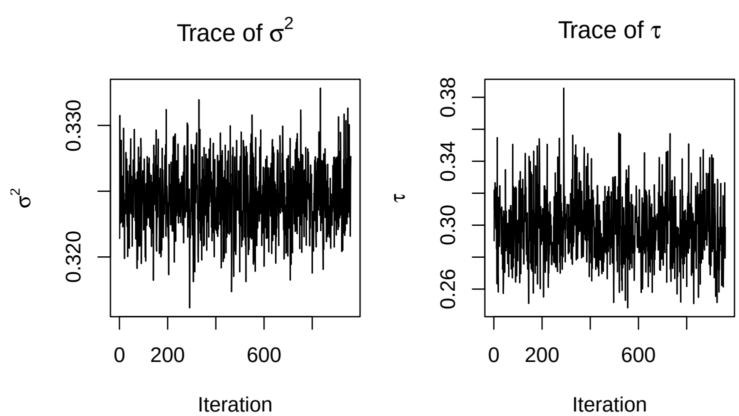

Figure 3: Post-burn-in trace plots for the variance components.

Autocorrelation functions for the variance components are reported in Figure 4. For both σ^2 and τ, autocorrelation declines rapidly after the first lag and remains close to zero for subsequent lags, suggesting satisfactory mixing.

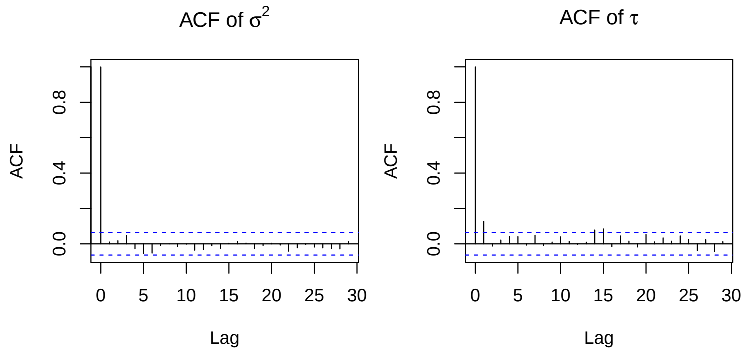

Figure 4: Post-burn-in autocorrelation functions for the variance components.

Autocorrelation functions were also inspected for representative regression coefficients and station-specific effects, selected based on their effective sample size to capture parameters with low, median, and high mixing efficiency. While some persistence is observed at low lags for a subset of parameters, particularly among station effects, autocorrelation decays steadily and remains within acceptable bounds. This pattern is consistent with the effective sample size results, which indicate some heterogeneity in mixing across parameters but no evidence of pathological behavior.

The Geweke diagnostic further supports convergence. Instead of reporting individual diagnostics for each regression coefficient and station effect, Table 2 summarizes the Geweke z-scores by parameter block. All reported values lie within the conventional acceptance interval $[-2, 2]$, providing no evidence against convergence.

For the regression coefficients, the Geweke z-scores range from -1.731 to 0.223, with an average value of -0.671. Similarly, for the station-specific effects, the values range from -0.777 to 1.166, with an average value of -0.037. These results indicate no systematic deviation from stationarity across either block.

The variance components show z-scores close to zero, equal to -0.649 for σ^2 and -0.200 for τ, further supporting convergence of the global parameters.

Table 2: Summary of Geweke diagnostic z-scores by parameter block.

Parameter block	Min	Mean	Max
β	-1.731	-0.671	0.223
α	-0.777	-0.037	1.166
σ^2	-0.649	-0.649	-0.649
τ	-0.200	-0.200	-0.200

Overall, the trace plots, autocorrelation functions, and Geweke diagnostics consistently indicate that the sampler mixes satisfactorily and provides reliable posterior draws for subsequent inference.

4.2 Posterior inference

The posterior estimates shown in Table 3 provide a clear characterization of the main sources of variation in PM2.5 concentrations.

The reported credible intervals are 95% highest posterior density (HPD) intervals. Effects whose intervals exclude zero are interpreted as credibly different from the baseline category.

The intercept represents the expected value of the log-transformed outcome for the reference category (January, Monday) at an average station. Its posterior mean is approximately 2.60, corresponding to a baseline PM2.5 level of about 13.5 on the original scale.

In addition, the posterior estimate of the residual standard deviation exceeds the group-level standard deviation, indicating that within-station variability is larger than persistent cross-station heterogeneity. In other words, day-to-day fluctuations in PM2.5 concentrations within the same monitoring site appear to be more substantial than the average differences in baseline pollution levels across stations.

At the same time, the group-level variation remains clearly non-negligible. This implies that, even after controlling for temporal effects, monitoring stations still differ in their structural pollution profiles. Such differences are consistent with the presence of persistent local factors, such as urban density, traffic exposure, industrial activity, and topographical or meteorological conditions.

From a modeling perspective, this result supports the use of a hierarchical specification. Although most of the variation is driven by within-station dynamics, the station-level random effects capture an important component of systematic heterogeneity that would be overlooked in a fully pooled model.

4.2.1 Temporal effects

The temporal coefficients reported in Table 3 are now interpreted in detail. A strong seasonal pattern emerges, as relative to January, nearly all months exhibit negative coefficients, indicating substantially lower pollution levels throughout most of the year.

The magnitude of these effects is substantively large: spring, summer, and early autumn months are associated with reductions often exceeding 30–50%. December and February appear indistinguishable from January in posterior terms, suggesting a return to high pollution levels during winter.

This pattern is consistent with well-known environmental mechanisms, including increased heating emissions and less favorable atmospheric dispersion conditions during colder months.

For compactness, Table 3 reports only the intercept and monthly effects. The omitted day-of-week coefficients are small and their 95% HPD intervals include zero, suggesting no systematic weekly cycle once seasonal and station-level heterogeneity are accounted for.

4.2.2 Spatial effects

Figure 5 displays the geographical distribution of monitoring stations, with colors representing the posterior mean of the station-specific random effects.

Table 3: Posterior means, 95% HPD intervals, and approximate percentage effects relative to the baseline category.

Variable	Mean	95% HPD interval	Effect
Intercept (Jan, Mon)	2.603	[2.503, 2.699]	Baseline (~13.5)
February	-0.016	[-0.103, 0.064]	-1.6%
March	-0.362	[-0.447, -0.277]	-30.4%
April	-0.615	[-0.698, -0.531]	-45.9%
May	-0.742	[-0.828, -0.657]	-52.4%
June	-0.119	[-0.21, -0.04]	-11.3%
July	-0.598	[-0.681, -0.506]	-45.0%
August	-0.425	[-0.504, -0.341]	-34.6%
September	-0.800	[-0.885, -0.71]	-55.0%
October	-0.605	[-0.688, -0.518]	-45.4%
November	-0.381	[-0.468, -0.299]	-31.7%
December	-0.068	[-0.152, 0.016]	+7.0%

The analysis is exploratory and relies on latitude-based macro-area groups rather than on a formal spatial model. Specifically, macro-areas are defined using latitude thresholds at 44° and 41.5°. Positive values indicate structurally higher PM2.5 levels after controlling for temporal effects.

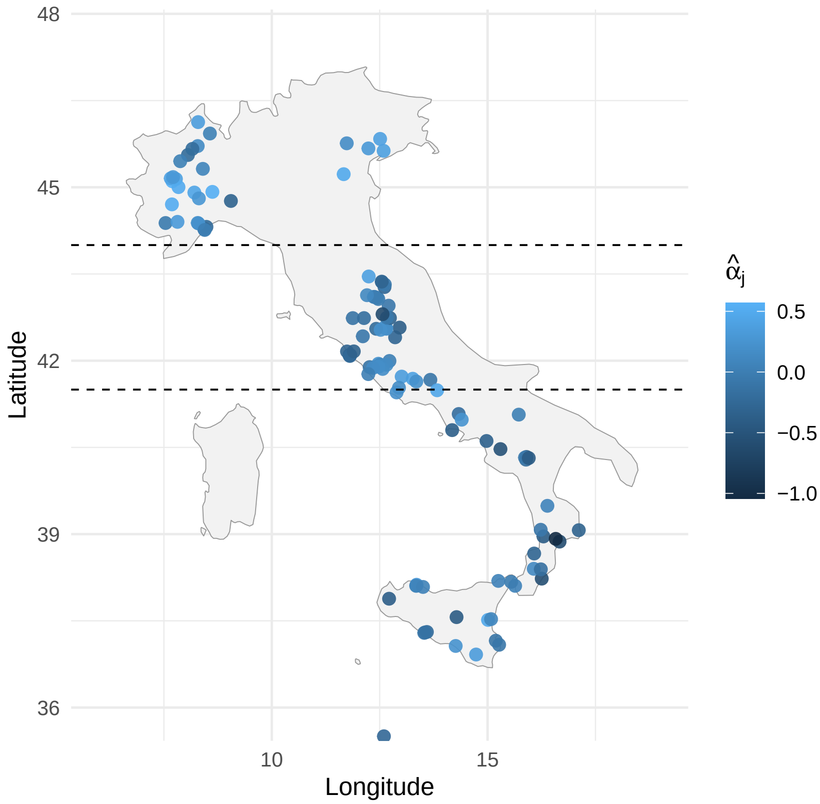

Figure 5: Spatial distribution of posterior mean station effects across Italy.

The spatial distribution reveals a clear North–South gradient in the station-level effects. Stations located in northern Italy tend to exhibit positive random effects, whereas southern stations display predominantly negative values, with central regions lying in between.

Table 4: Posterior summaries of average station effects by macro-area.

Area	Mean	95% HPD interval	Pr(>0)
North	0.195	[0.131, 0.246]	1.000
Center	-0.016	[-0.071, 0.042]	0.303
South	-0.113	[-0.169, -0.055]	0.000
North - South	0.308	[0.29, 0.328]	1.000
North - Center	0.210	[0.196, 0.229]	1.000

This pattern is confirmed by a posterior comparison of average station effects across macro-areas. The North–South difference is estimated at approximately 0.31 on the log scale, with a 95% credible interval entirely above zero. The posterior probability that the North exhibits higher baseline pollution than the South is effectively equal to one.

On the original scale, this corresponds to roughly a 36% higher structural level of $PM_{2.5}$ in the North relative to the South.

This evidence is consistent with well-documented geographical differences in pollution dynamics, including higher industrial intensity, traffic exposure, and less favorable atmospheric dispersion conditions in northern regions. At the same time, the dispersion of the station effects highlights substantial within-region heterogeneity, suggesting that local factors remain important determinants of air quality.

Overall, these results highlight the coexistence of broad geographical gradients and substantial local heterogeneity in $PM_{2.5}$ dynamics.

4.3 Posterior predictive checks

Posterior predictive checks were performed to assess model adequacy. Replicated datasets were generated from the posterior distribution and compared to the observed data.

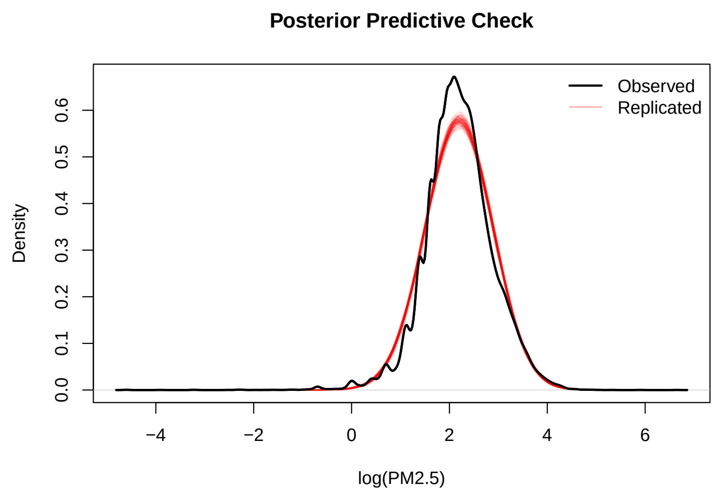

Figure 6: Posterior predictive check for the log-transformed PM2.5 outcome.

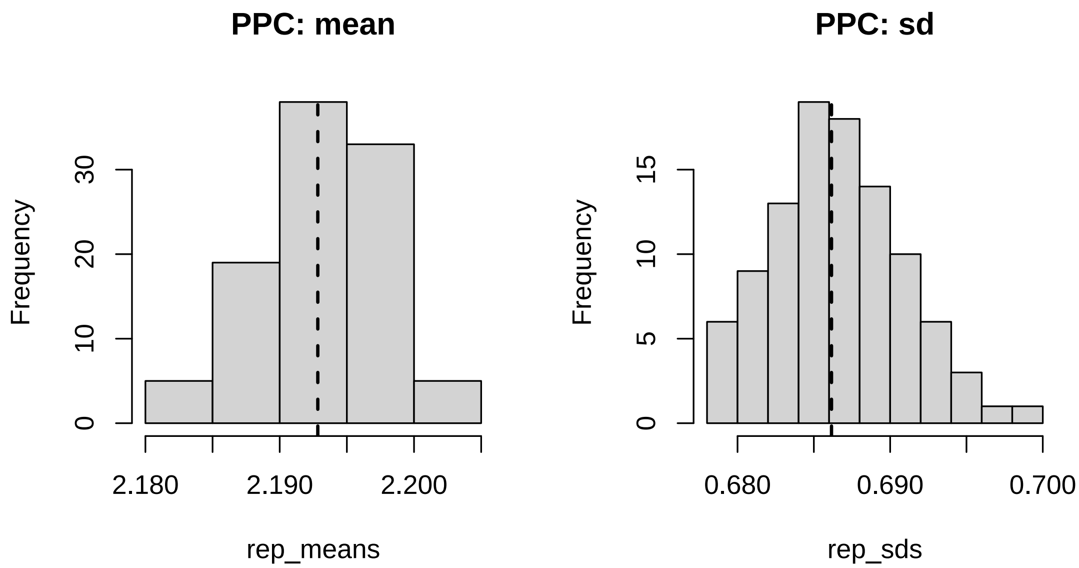

Figure 7: Posterior predictive checks based on summary statistics. The observed mean and standard deviation fall well within the distribution of replicated values.

The distribution of replicated means and standard deviations is consistent with the observed values (see Figures 6 and 7), indicating that the model accurately captures the first two moments of the data.

Furthermore, the posterior predictive density aligns well with the empirical distribution of log-transformed PM2.5, with only minor discrepancies in the peak and tail regions. These differences are consistent with the Gaussian assumption and do not indicate major model misspecification.

Robustness checks based on alternative prior calibrations, reported in Section 3.4.1, confirm that the main posterior conclusions are stable.

5 Conclusion

This paper develops a Bayesian hierarchical model for daily PM2.5 concentrations, explicitly accounting for heterogeneity across monitoring stations and the unbalanced structure of the panel.

The Metropolis-within-Gibbs sampler exhibits satisfactory convergence behavior, as supported by multiple diagnostic tools. Posterior inference appears stable with respect to prior specification, and posterior predictive checks indicate that the model is able to capture the main features of the observed data.

From a substantive perspective, the results highlight a pronounced seasonal pattern, with substantially higher PM2.5 levels during winter months. This finding reinforces the relevance of seasonally targeted environmental policies, particularly those addressing emissions related to heating systems and increased urban activity. By contrast, the absence of a clear weekly cycle suggests that short-term policy interventions (e.g., day-specific restrictions) may be insufficient unless embedded within broader, structural measures.

In addition, the presence of persistent differences across monitoring stations, together with the identified North–South gradient, points to the importance of geographically differentiated policy approaches. These results are consistent with the idea that local environmental conditions, urban structure, and regional economic activity jointly shape air pollution dynamics.

Despite these strengths, the analysis is subject to several limitations. First, the study focuses on a single year (2025), which prevents the investigation of longer-term trends and inter-annual variability. Extending the analysis to multiple years would allow for a more robust assessment of persistence, structural changes, and the potential impact of policy interventions over time.

Second, the spatial coverage of the dataset is limited to 116 monitoring stations and is not fully representative of the entire Italian territory. Some regions are underrepresented, which may introduce sampling bias and limit the external validity of the results, particularly with respect to the estimated spatial gradient.

Third, while the hierarchical structure partially mitigates data imbalance, the irregular temporal coverage across stations may still affect inference, especially for locations with substantial missing periods.

Finally, the current specification does not explicitly model spatial dependence across stations, nor does it incorporate meteorological covariates (e.g., temperature, wind, humidity), which are known to play a crucial role in pollution dynamics.

These limitations suggest several directions for future research. Extending the dataset across multiple years and improving spatial coverage would enhance the robustness of the empirical findings. Incorporating spatial correlation structures, such as Gaussian processes or conditional autoregressive (CAR) models, would allow for a more realistic representation of geographic dependence. Additionally, enriching the model with meteorological and environmental covariates could improve both explanatory power and predictive performance.

More flexible likelihood specifications (e.g., heavy-tailed or skewed distributions) may also be considered to better capture extreme pollution episodes. Finally, the framework could be extended to a fully spatio-temporal hierarchical model, providing a unified approach to jointly model temporal dynamics and spatial interactions.

Overall, the hierarchical Bayesian framework proves to be a flexible and effective tool for the analysis of air pollution data, offering both methodological rigor and policy-relevant insights.